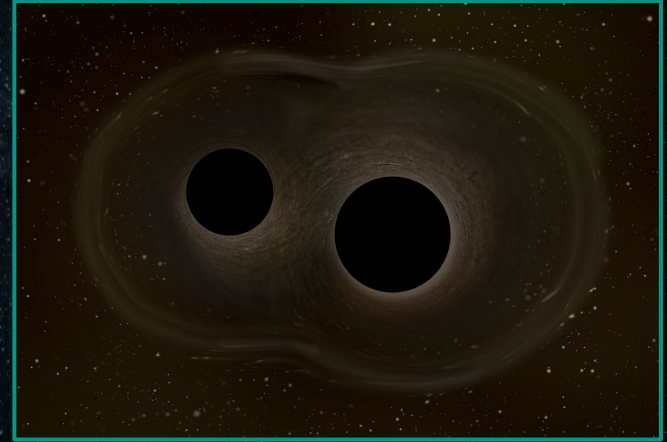


Correcting Compact Binary Inspiral Gravitational Waveform Uncertainty in Bayesian Inference Parameter Estimation

Ryan Johnson, Dr. Jocelyn Read

Compact Binary Inspiral Gravitational Waves

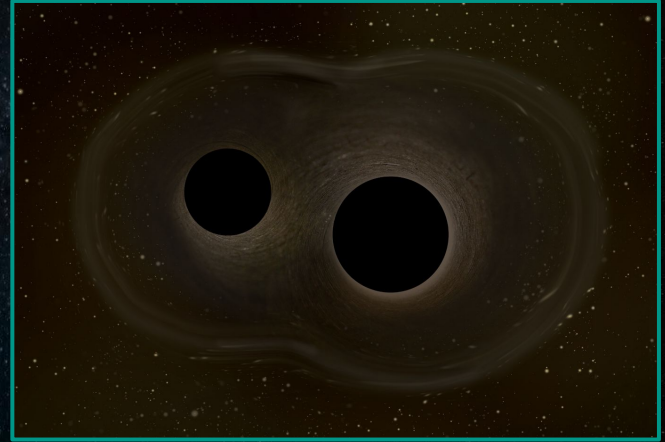
- Produced by binary black holes and neutron stars
 - Focus on black holes
- Measured by interferometers
 - LIGO, Virgo, Kagra
- Signals received as gravitational wave strain, h
- Two domains
 - Time domain
 - Frequency (Fourier) domain



Compact Binary Inspiral Gravitational Waves

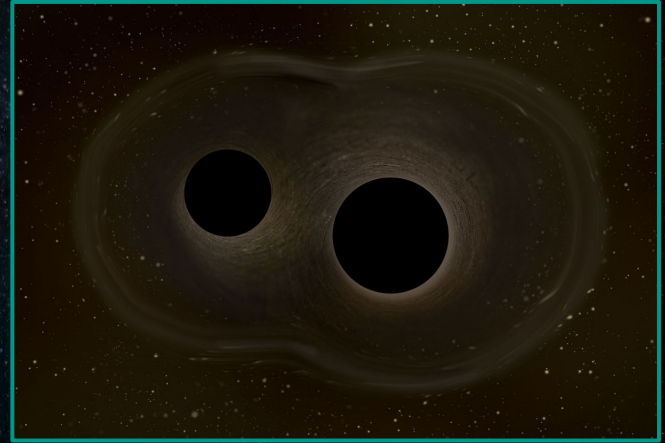
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$$h(t) = A(t)e^{i\psi(t)}$$



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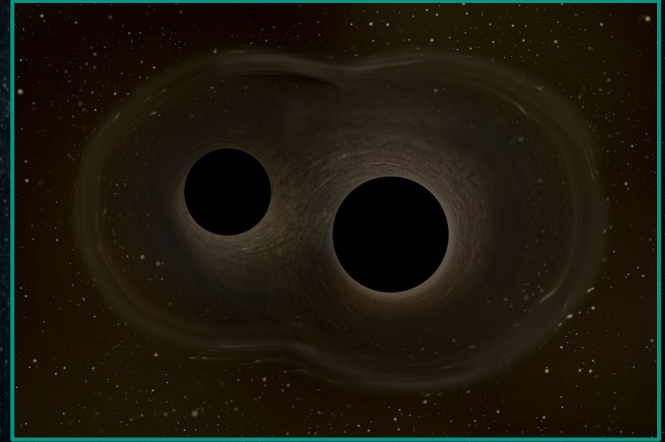
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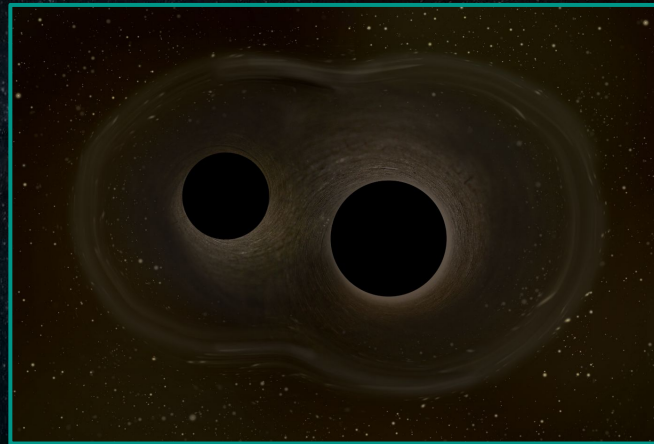
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$$h(t) = A(t)e^{i\psi(t)}$$

$$\tilde{h}(f) = \mathcal{A}(f)e^{i\phi(f)}$$

$$\mathcal{A}(f) = \sqrt{\tilde{h}^*(f)\tilde{h}(f)}$$

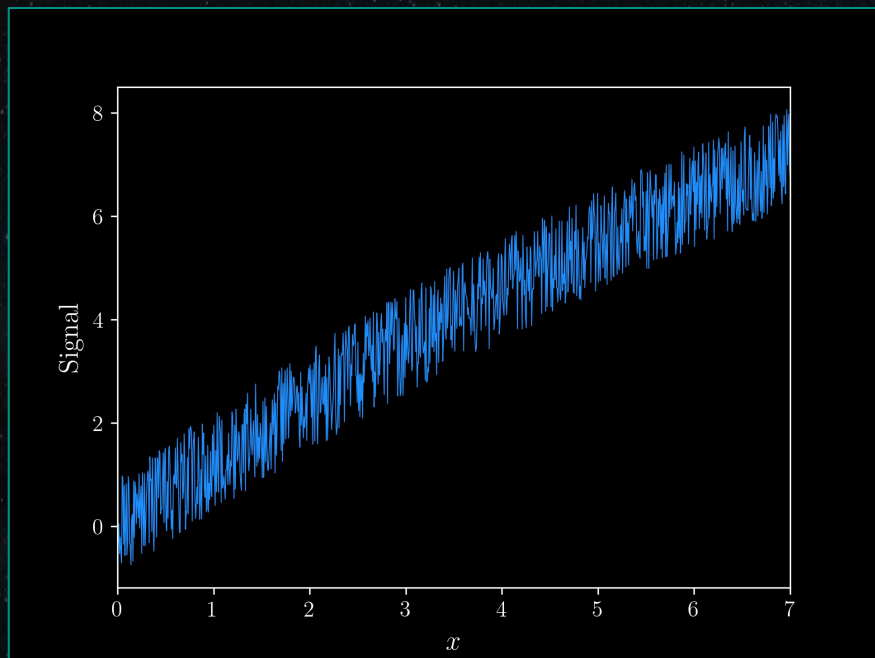
$$\phi(f) = \arctan\left(\frac{\text{Im}[\tilde{h}(f)]}{\text{Re}[\tilde{h}(f)]}\right)$$

What is Parameter Estimation?

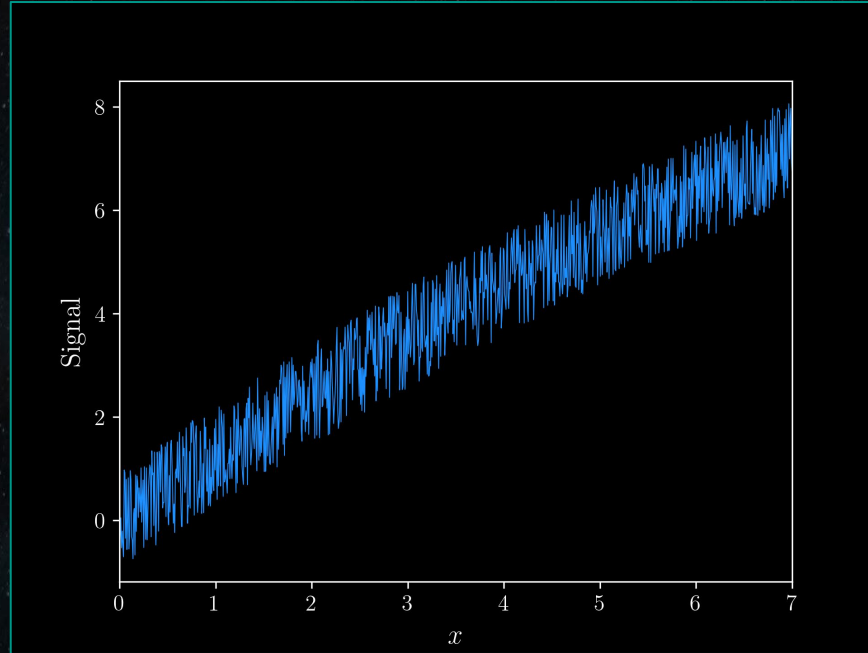
- Method of determining the source properties of a gravitational-wave signal
- Fitting a gravitational-wave model to a signal
 - Model parameters that best match the signal are taken to be the source parameters that created the signal

What is Waveform Uncertainty?

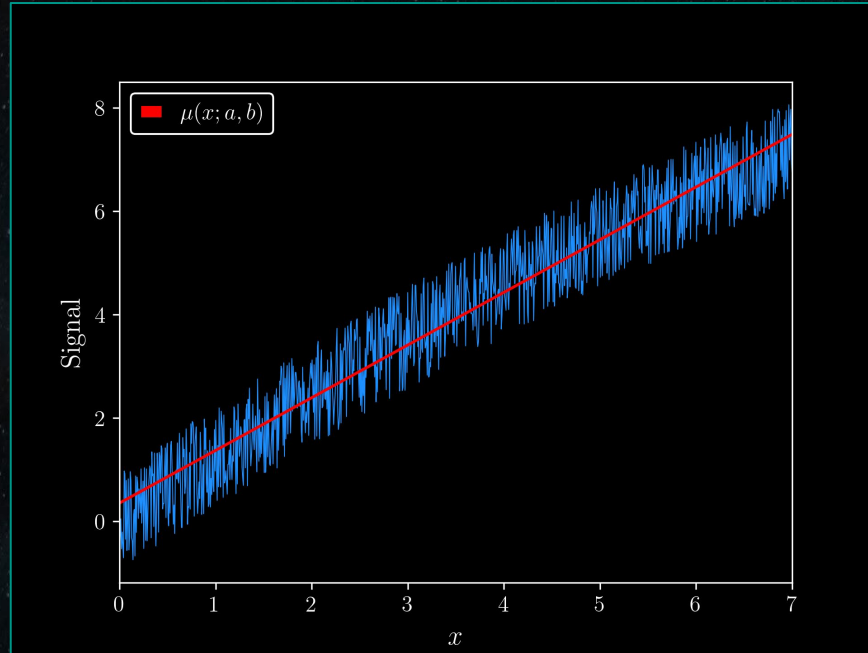
- Uncertainty in the output of a parameter estimation run
- Caused by models being unable to correctly fit to a gravitational-wave signal
 - Waveform models are not perfect



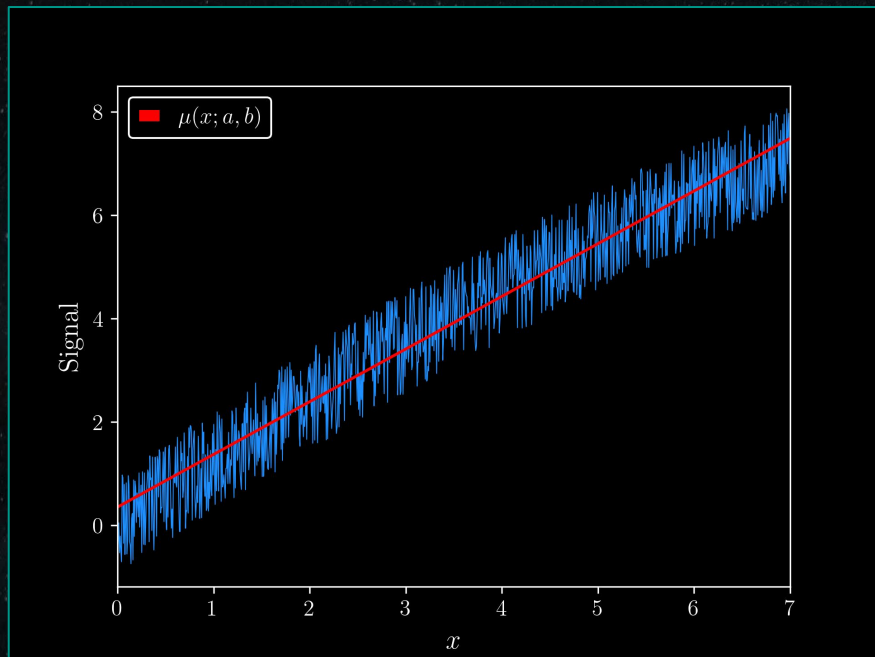
$$\mu(x; a, b) = ax + b$$



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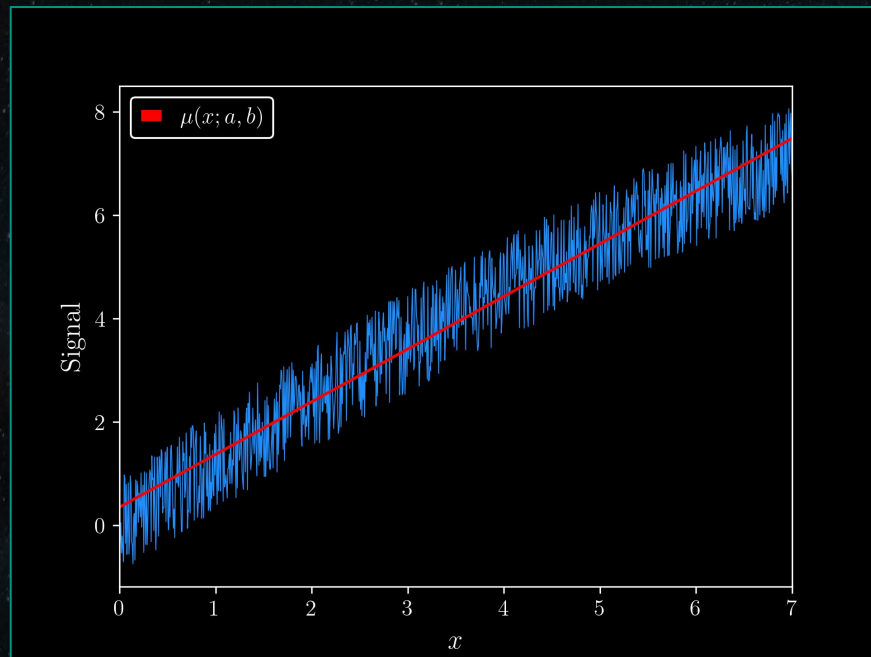


$$\mu(x; a, b) = ax + b$$



$$a = 1.0195$$
$$b = 0.3504$$

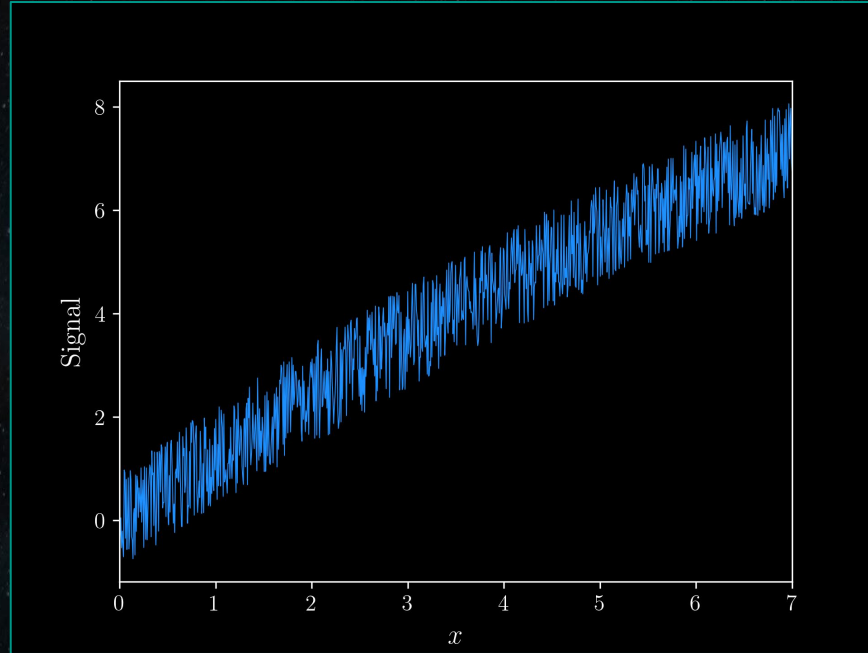
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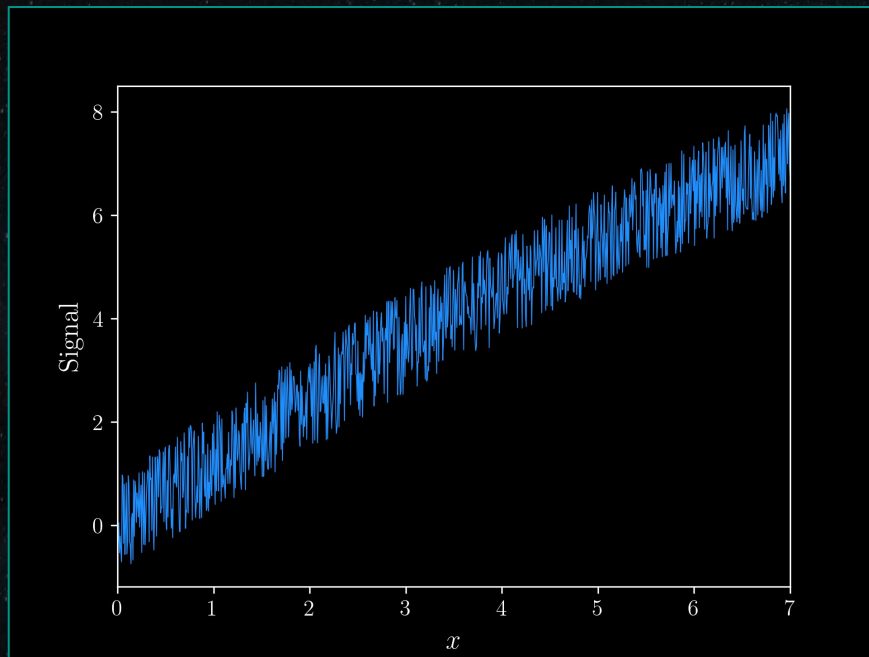
$$\begin{aligned} a &= 1.0195 \\ b &= 0.3504 \end{aligned}$$

?

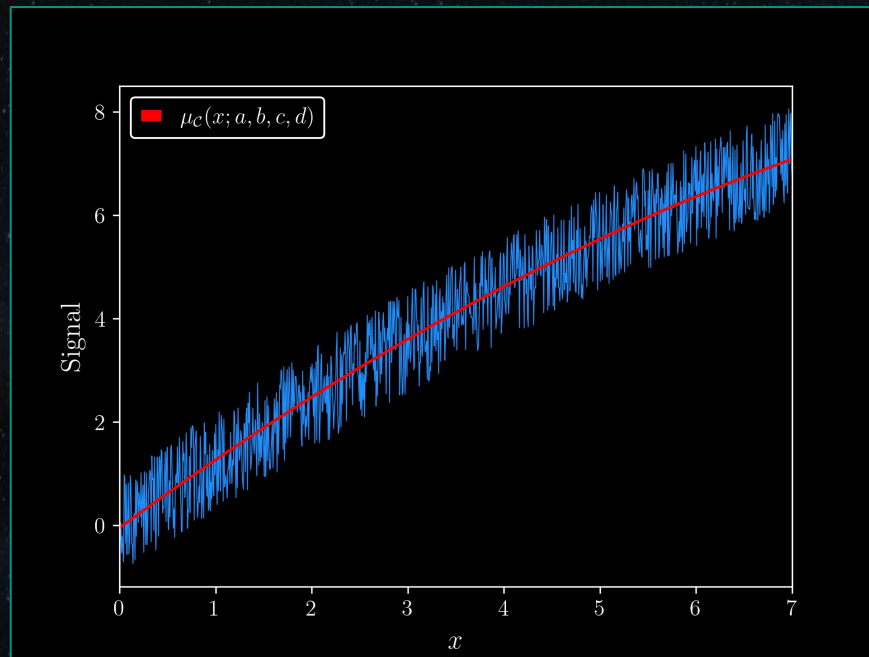
$$\mu_{\mathcal{C}}(x; a, b, c, d) = ax + b + cx^d$$



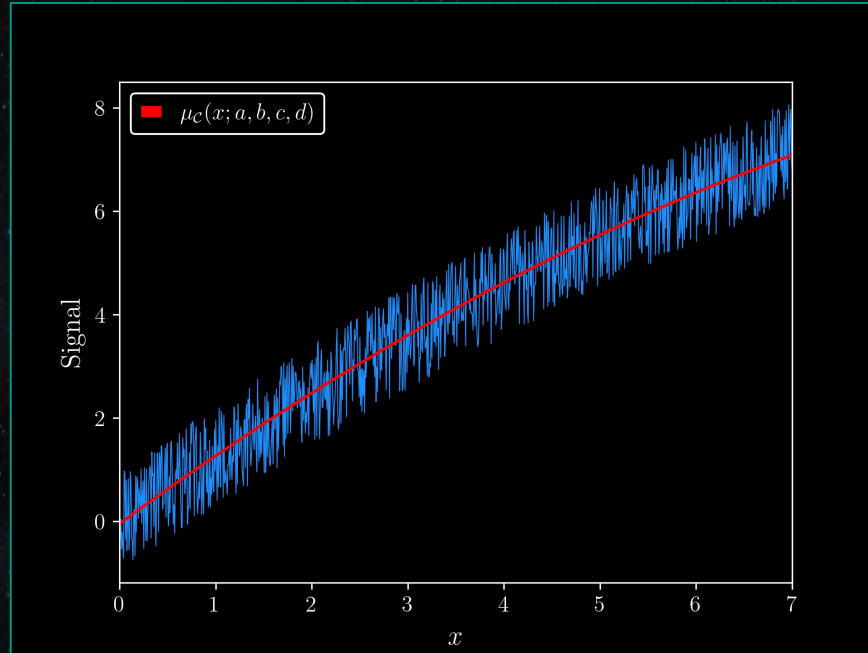
$$\mu_{\mathcal{C}}(x; a, b, c, d) = ax + b + \underline{cx^d}$$



$$\mu_C(x; a, b, c, d) = ax + b + cx^d$$



$$\mu_C(x; a, b, c, d) = ax + b + cx^d$$



$$\begin{aligned} a &= 1.3644 \\ b &= -0.0515 \end{aligned}$$

ϑ	ϑ^{true}	$\vartheta_{\emptyset}^{\text{ml}}$	$\vartheta_{\mathcal{C}}^{\text{ml}}$
a	1.3	1.0195	1.3644
b	0.002	0.3504	-0.0515

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a	1.3	1.0195	1.3644
b	0.002	0.3504	-0.0515

To Apply to Gravitational Waves:

1. Define a new waveform model with correction parameters
2. Place boundary conditions on the correction parameters
3. Define prior distributions for each correction parameter
4. Perform parameter estimation tests:
 - a. NR Consistency Check
 - b. Injected Phase Error Test

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New Waveform Model

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New Waveform Model

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$$\tilde{h}_{\text{true}}(f) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}(f)) \exp(i\Delta\phi(f))$$

$$\Delta\mathcal{A}(f) = \frac{\mathcal{A}_{\text{true}}(f)}{\mathcal{A}_{\text{model}}(f)} - 1 \quad \Delta\phi(f) = \phi_{\text{true}}(f) - \phi_{\text{model}}(f)$$

New Waveform Model (cont.)

- We need functional forms for the corrections
 - Cannot assume any structure

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$$\Delta\mathcal{A}(f) \rightarrow \Delta\mathcal{A}_s(f; \{f_k, \alpha_k\})$$

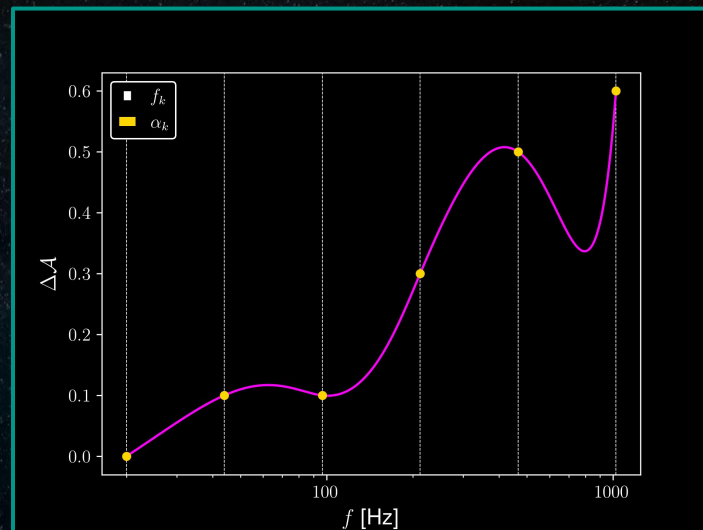
$$\Delta\phi(f) \rightarrow \Delta\phi_s(f; \{f_k, \varphi_k\})$$

New Waveform Model (cont.)

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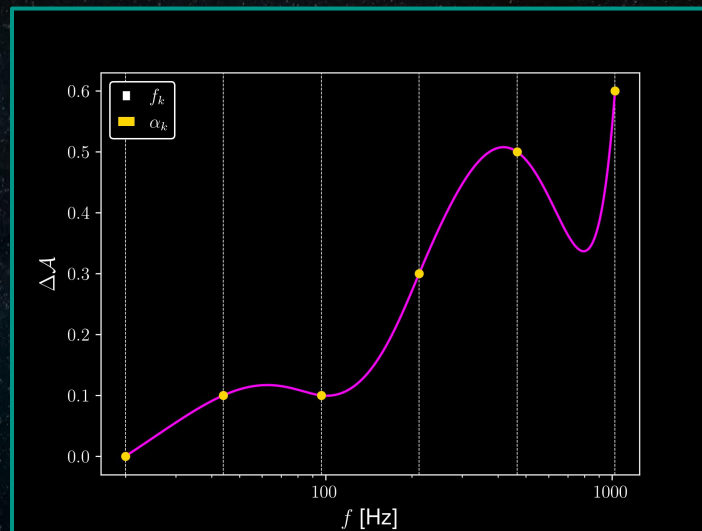


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$$\Delta\phi(f) \rightarrow \Delta\phi_s(f; \{f_k, \varphi_k\})$$



$$\mu_c(f; \vartheta, \alpha, \varphi) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}_s(f; \{f_k, \alpha_k\})) \exp(i\Delta\phi_s(f; \{f_k, \varphi_k\}))$$

Frequency Node Boundary Conditions

- Where should the frequency nodes be placed?
- Assume insignificant waveform uncertainty in the early inspiral regime

Lower Bound

- Frequency at the end of the early inspiral regime

Upper Bound

- Frequency when the tangential velocity reaches lightspeed

Frequency Node Boundary Conditions (cont.)

$$M f_{\text{IM}} = 0.018 \leftarrow \text{Khan et. al. 2015}$$

$$M f_{\text{light}} = \pi^{-1} \leftarrow \text{Classical Mechanics}$$

Frequency Node Boundary Conditions (cont.)

$$M f_{\text{IM}} = 0.018$$

$$M f_{\text{light}} = \pi^{-1}$$

$$\xi \equiv \frac{GMf}{c^3}$$

Frequency Node Boundary Conditions (cont.)

$$\begin{array}{l} M f_{\text{IM}} = 0.018 \\ M f_{\text{light}} = \pi^{-1} \end{array} \xrightarrow{\text{red}} \xi \equiv \frac{GMf}{c^3} \xrightarrow{\text{red}} \begin{array}{l} \xi_{\text{IM}} = 0.018 \\ \xi_{\text{light}} = \pi^{-1} \end{array}$$

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$$\{\xi_k\} = \frac{GM}{c^3} \{f_k\}$$

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$$\{\xi_k\} = \frac{GM}{c^3} \{f_k\}$$

$$\begin{cases} \alpha_k, \varphi_k = 0 & \xi \leq 0.018 \\ \alpha_k, \varphi_k = \text{constant} & \xi > \pi^{-1} \end{cases}$$

Waveform Correction Prior

- What values can each correction parameter take on?
- What should the probability distribution look like?

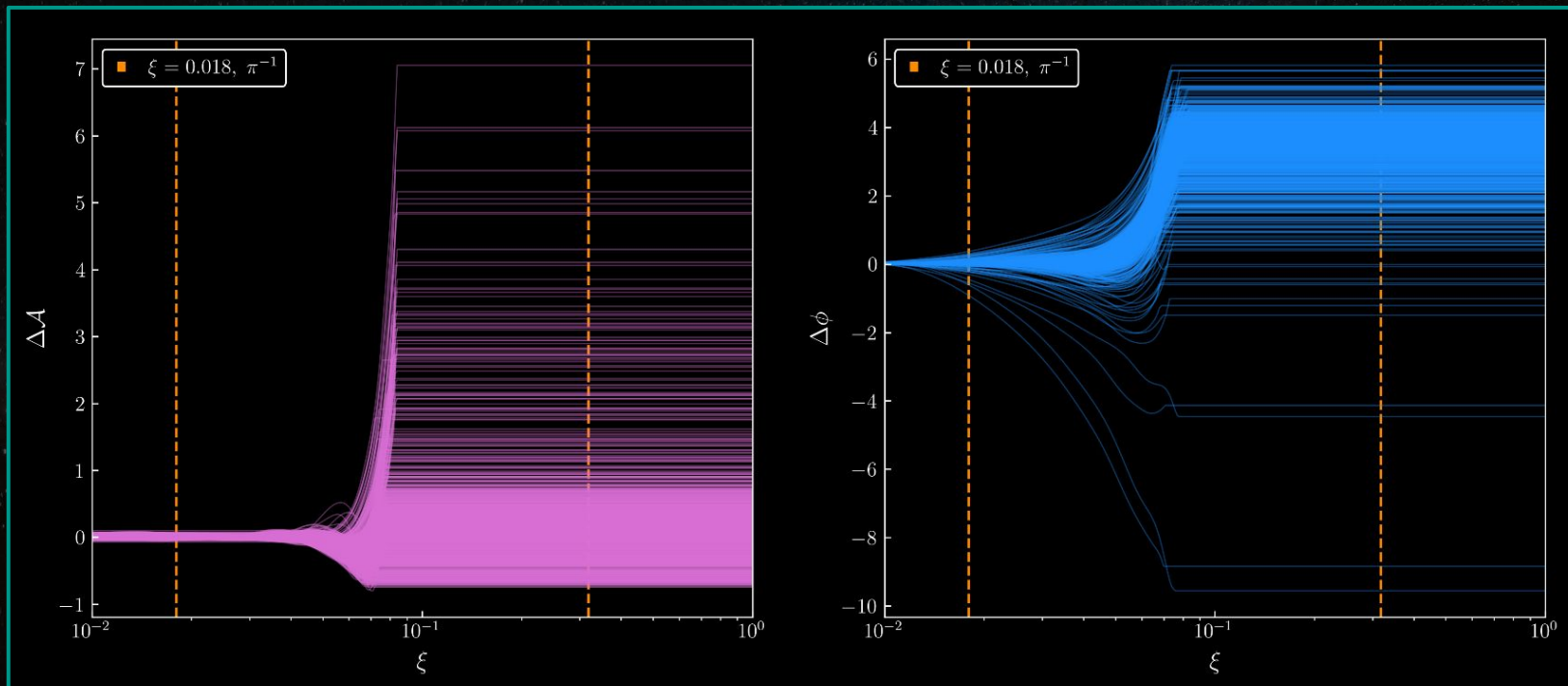
We look to the amplitude and phase differences between two waveform model families: **IMR** and **EOB**

- Work in mass-weighted dimensionless frequency to ensure boundary conditions

Waveform Correction Prior (cont.)

$$\Delta\mathcal{A}_\mu(\xi; \vartheta) = \frac{\mathcal{A}_{\text{EOB}}(\xi; \vartheta)}{\mathcal{A}_{\text{IMR}}(\xi; \vartheta)} - 1$$

$$\Delta\phi_\mu(\xi; \vartheta) = \phi_{\text{EOB}}(\xi; \vartheta) - \phi_{\text{IMR}}(\xi; \vartheta) - 2\pi f \Delta t_c - \Delta\phi_c$$



Waveform Correction Prior (cont.)

$$\overline{\Delta\mathcal{A}_\mu}(\xi) = \frac{\sum_{i=1}^N (\Delta\mathcal{A}_\mu(\xi; \vartheta_i))}{N} \quad \delta\mathcal{A}_\mu(\xi) = \sqrt{\frac{\sum_{i=1}^N (\Delta\mathcal{A}_\mu(\xi; \vartheta_i) - \overline{\Delta\mathcal{A}_\mu}(\xi))^2}{N}}$$

$$\overline{\Delta\phi_\mu}(\xi) = \frac{\sum_{i=1}^N (\Delta\phi_\mu(\xi; \vartheta_i))}{N} \quad \delta\phi_\mu(\xi) = \sqrt{\frac{\sum_{i=1}^N (\Delta\phi_\mu(\xi; \vartheta_i) - \overline{\Delta\phi_\mu}(\xi))^2}{N}}$$

Waveform Correction Prior (cont.)

We want zero waveform correction to be the most likely

- Set the mean of the distributions to zero

Waveform Correction Prior (cont.)

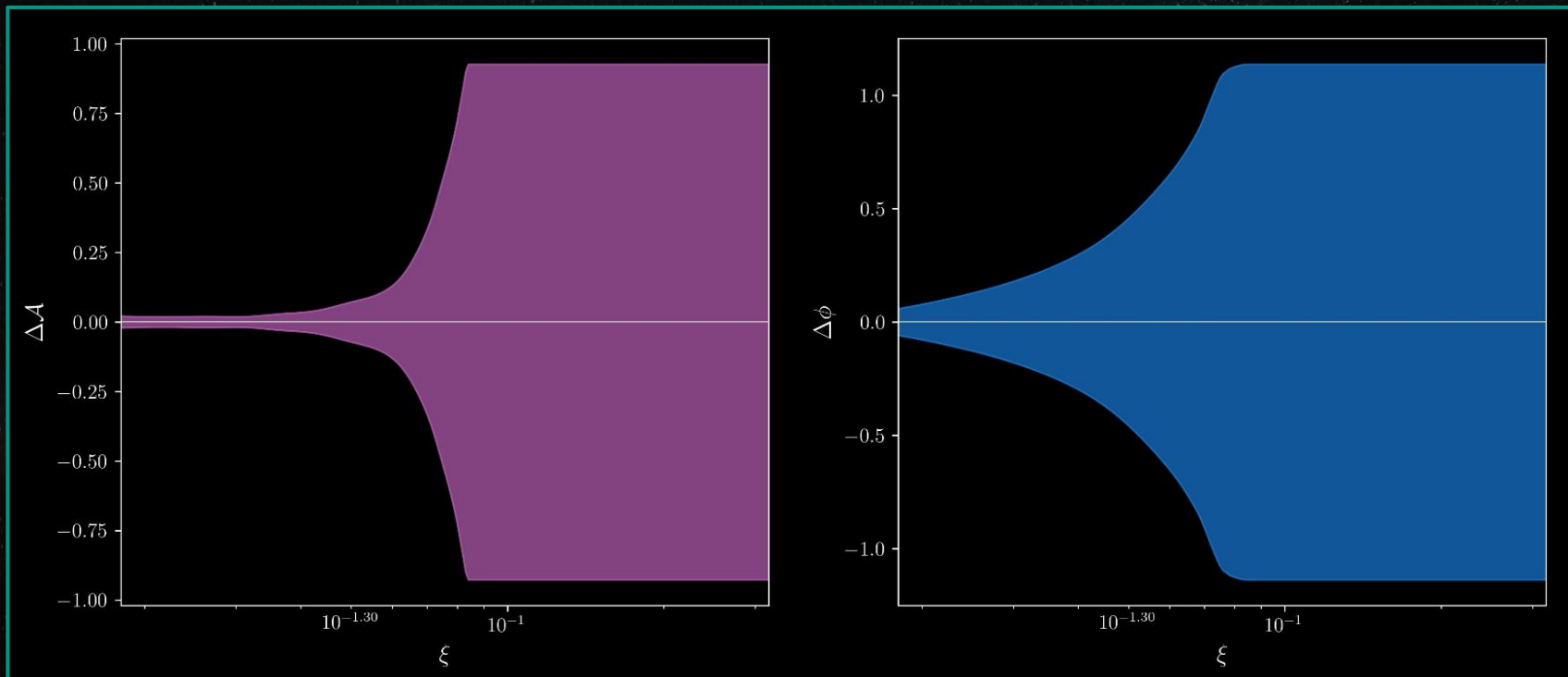
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- Set the mean of the distributions to zero

$$\Pi(\alpha_k) = \mathcal{N}(0, \delta \mathcal{A}_\mu(\xi_k))$$

$$\Pi(\varphi_k) = \mathcal{N}(0, \delta \phi_\mu(\xi_k))$$

Waveform Correction Prior (cont.)



Theory Summary

1. We introduced a corrected waveform model

$$\mu_c(f; \vartheta, \alpha, \varphi) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}_s(f; \{f_k, \alpha_k\})) \exp(i\Delta\phi_s(f; \{f_k, \varphi_k\}))$$

2. We chose where the waveform corrections should be in the waveform

$$\begin{cases} \alpha_k, \varphi_k = 0 & \xi \leq 0.018 \\ \alpha_k, \varphi_k = \text{constant} & \xi > \pi^{-1} \end{cases}$$

3. We defined prior distributions for each waveform correction parameter

$$\Pi(\alpha_k) = \mathcal{N}(0, \delta\mathcal{A}_\mu(\xi_k)) \quad \Pi(\varphi_k) = \mathcal{N}(0, \delta\phi_\mu(\xi_k))$$

Test Setup

- Choose GW190521-like binary black hole source parameters
- Choose **IMRPhenomPv2** waveform model
- Perform parameter estimation with **bilby** (Ashton et. al. 2019)

$$\vartheta^{\text{inj}} = \left\{ \begin{array}{l} M = 150.81 \text{ M}_{\odot} \\ \mathcal{M}_c = 65.10 \text{ M}_{\odot} \\ q = 0.79 \\ \chi_{\text{eff}} = -0.08 \\ \chi_p = 0.61 \\ D_L = 637.0 \text{ Mpc} \\ \theta_{JN} = 0.85 \end{array} \right.$$

Test: NR Consistency

- If a standard waveform model is used to create a signal, the corrected waveform model should have nothing to correct and will behave as a standard waveform model.
- Analyze standard waveform with and without corrections
- Choose 6 frequency nodes/phase correction parameters

$$\varphi^{\text{inj}} = \begin{cases} \varphi_1 = 0 \\ \varphi_2 = 0 \\ \varphi_3 = 0 \\ \varphi_4 = 0 \\ \varphi_5 = 0 \\ \varphi_6 = 0 \end{cases}$$

Test: NR Consistency Results

ϑ	ϑ^{inj}	$P_{\emptyset}(\vartheta h)$	$P_{\mathcal{C}}(\vartheta h)$
$M \text{ [M}_{\odot}\text{]}$	150.81	$149.77^{+1.86}_{-1.95}$	$150.79^{+4.88}_{-4.97}$
$\mathcal{M}_c \text{ [M}_{\odot}\text{]}$	65.10	$64.90^{+0.86}_{-1.14}$	$65.00^{+2.27}_{-2.60}$
q	0.79	$0.86^{+0.09}_{-0.11}$	$0.80^{+0.13}_{-0.14}$
χ_{eff}	-0.08	$-0.08^{+0.03}_{-0.03}$	$-0.02^{+0.11}_{-0.12}$
χ_p	0.61	$0.63^{+0.10}_{-0.10}$	$0.51^{+0.17}_{-0.16}$
$D_L \text{ [Mpc]}$	637.00	$571.52^{+83.02}_{-49.62}$	$702.65^{+83.66}_{-90.33}$
θ_{JN}	0.85	$0.95^{+0.13}_{-0.22}$	$0.65^{+0.22}_{-0.22}$

Test: Injected Phase Error

- If there is a phase error present, an unmodified waveform model should show errors in the posteriors. The corrected waveform model should show reduced errors.
- Choose 6 frequency nodes/phase correction parameters
- Choose a random form for the phase error

$$\varphi^{\text{inj}} = \begin{cases} \varphi_1 = -0.1318 \\ \varphi_2 = 0.54134 \\ \varphi_3 = 0.99884 \\ \varphi_4 = 0.38220 \\ \varphi_5 = 0.54266 \\ \varphi_6 = 0.82875 \end{cases}$$

Test: Injected Phase Error Results

ϑ	ϑ^{inj}	$P_{\varnothing}(\vartheta h)$	$P_c(\vartheta h)$
$M \text{ [M}_{\odot}\text{]}$	150.81	$169.78^{+2.17}_{-2.13}$	$153.88^{+4.47}_{-4.56}$
$\mathcal{M}_c \text{ [M}_{\odot}\text{]}$	65.10	$73.33^{+1.02}_{-0.96}$	$66.28^{+2.08}_{-2.24}$
q	0.79	$0.83^{+0.09}_{-0.10}$	$0.82^{+0.12}_{-0.15}$
χ_{eff}	-0.08	$0.16^{+0.05}_{-0.07}$	$0.05^{+0.10}_{-0.11}$
χ_p	0.61	$0.89^{+0.06}_{-0.09}$	$0.56^{+0.16}_{-0.17}$
$D_L \text{ [Mpc]}$	637.00	$701.22^{+189.43}_{-115.48}$	$760.74^{+67.11}_{-74.65}$
θ_{JN}	0.85	$0.92^{+0.28}_{-0.59}$	$0.56^{+0.18}_{-0.18}$

Test: Injected Phase Error Results

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How was this work done?

- **WaveformUncertainty**
 - bilby-dependent Python package
 - 10 functions
 - Complete documentation
 - Ipython notebook tutorials



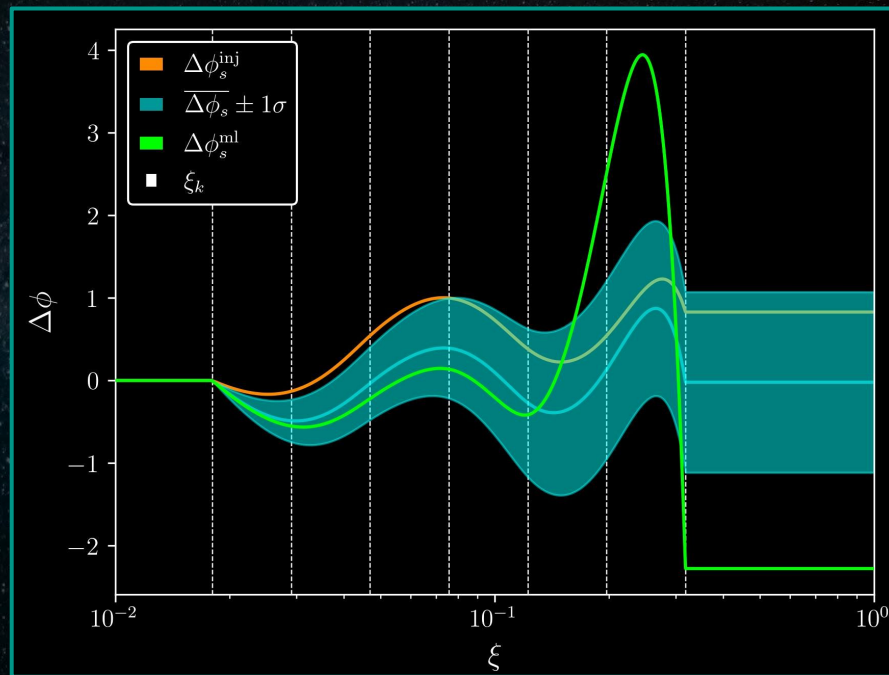
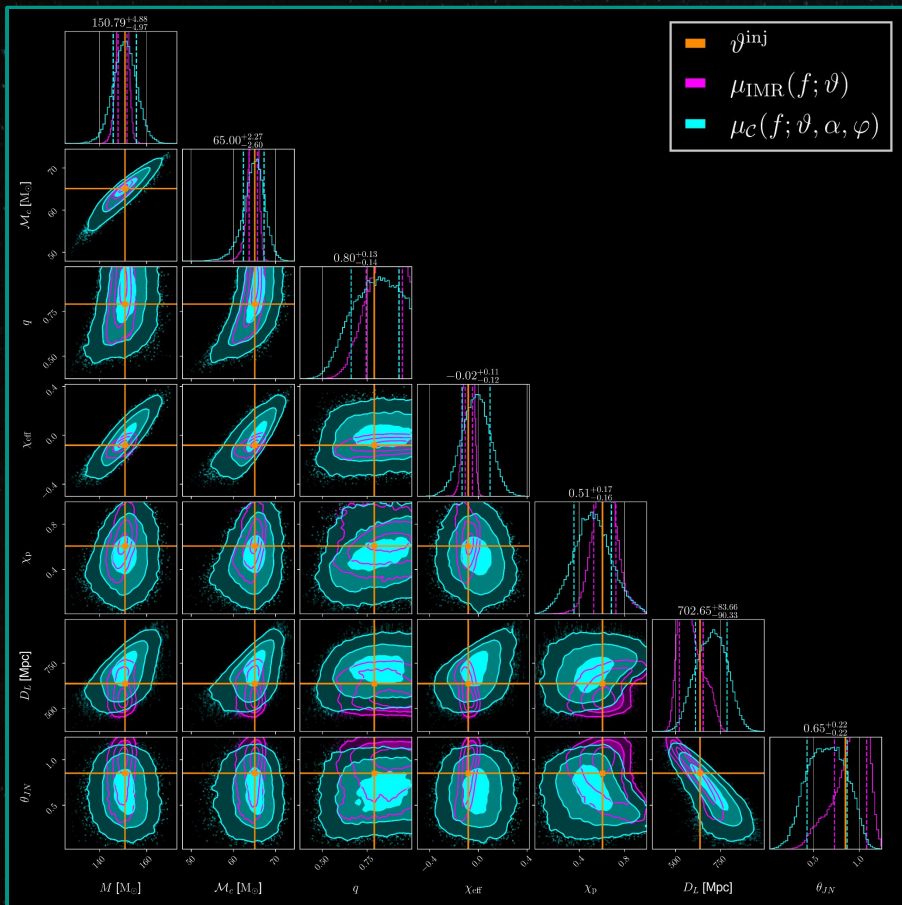
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Acknowledgements

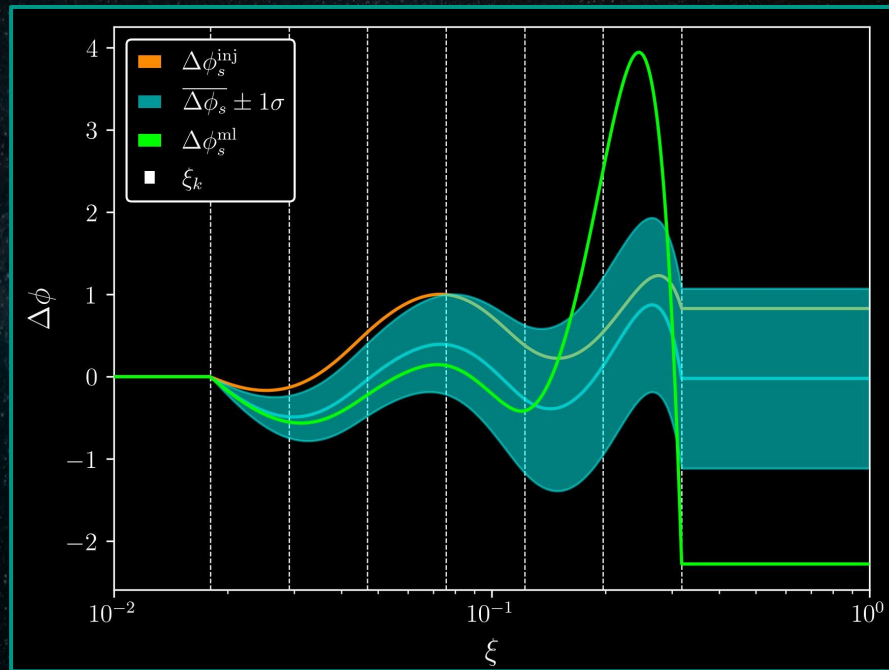
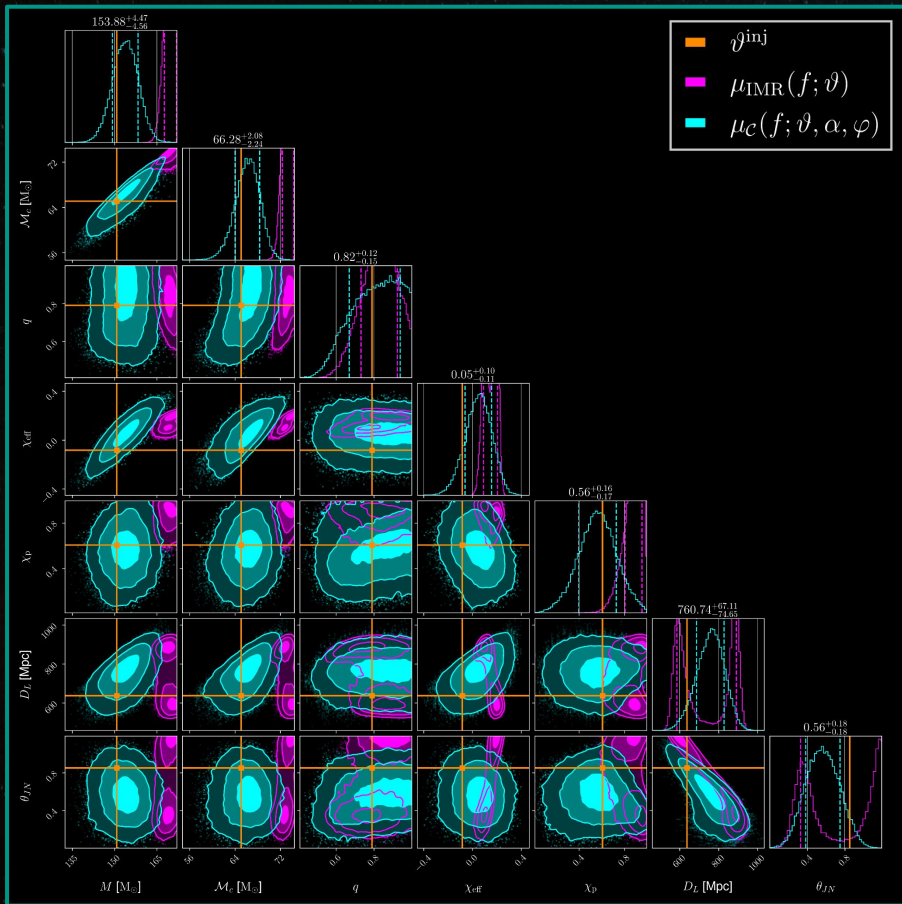
- Black Family Fellowship
- NSF PHYS-2409736
- Dr. Lami Suleiman
- Dr. Marceline Bonilla



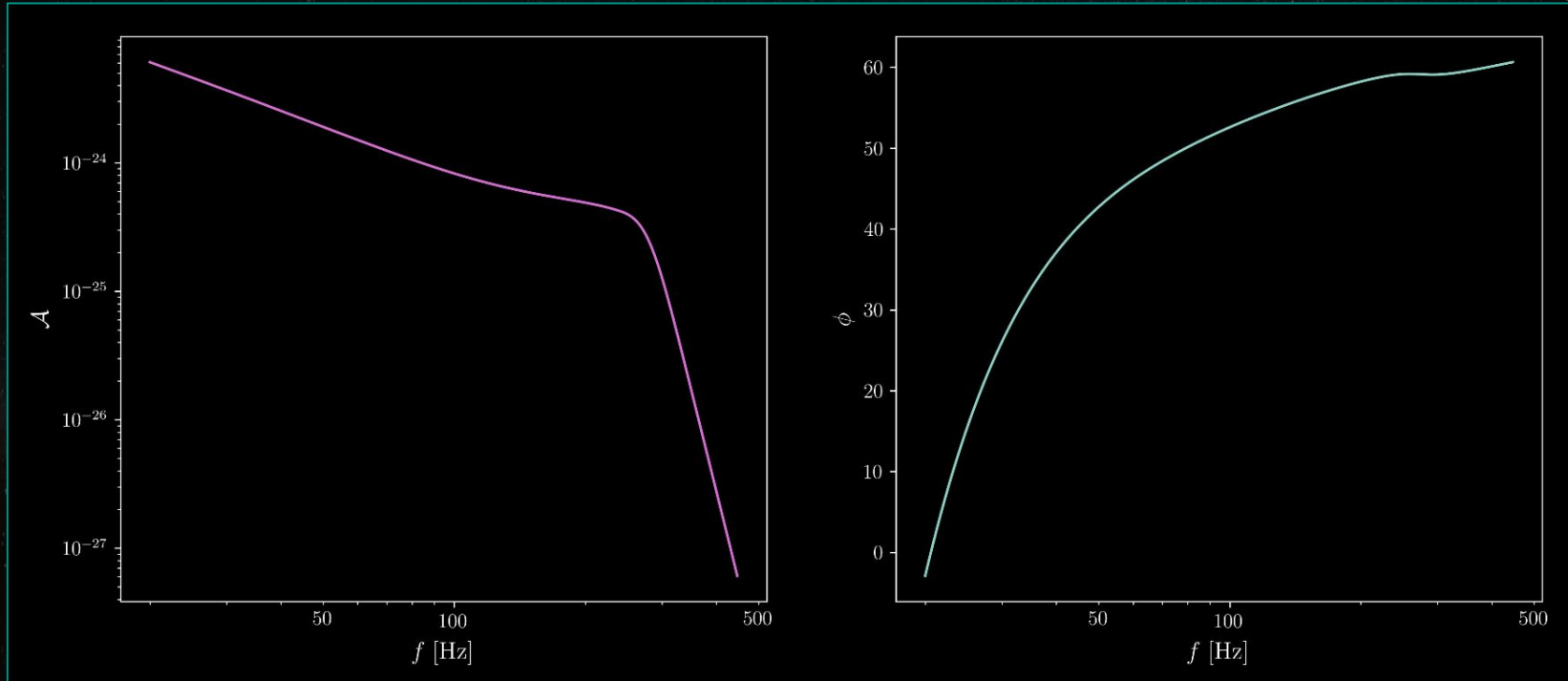
Test: NR Consistency Posterior



Test: Injected Phase Error Posterior



Frequency Domain Waveform



Total Mass Constraint

$$\frac{\pi^{-1}c^3}{Gf_{\text{high}}} < M < \frac{0.018c^3}{Gf_{\text{low}}}$$